



TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

Self-contained power supply coin battery module

SMK has introduced the industry's first energy-harvesting module designed to replace the CR2032 coin cell battery. The module integrates solar energy harvesting with Bluetooth low energy 5.3 for low-power communication, eliminating the need for battery replacement in remote controllers and sensors. Its compact design fits into existing coin cell battery slots, combining a battery, PV cell, communication circuit, and sensors within the same space. This enables both power generation and communication without modifying device enclosures. The module also supports wireless power transmission via an external antenna, making it usable in environments where solar power is limited. The company claims that it can be used in applications like automotive key FOBs, bicycle systems, IoT trackers, PC peripherals, and remote controllers.



SMK
<https://www.smkusa.com>

High-current relay for vehicles

Bosch has developed the HCR 100 high-current relay to meet the growing demand for high-current control in vehicles' electric systems. Designed for reliable switching of up to 100 amps, it surpasses previous relays limited to



75 amps. Typical applications include engine management, glow plugs, pre-heating systems, loading ramps, start-stop systems, and heated windshields. The HCR 100 suits various vehicles, including trucks, agricultural machinery, buses, campers, boats, and stationary applications. It features minimal contact resistance, operates at 100A at 85°C, and comes in 12V and 24V versions. Designed for temperatures from -40°C to +85°C, the relay has an electrical lifespan of up to 100,000 switching cycles.

Bosch
<https://www.bosch.in>

Infrared thermometer using optical filtering for induction hobs

Melexis have launched the MLX90617, an infrared thermometer designed for non-contact temperature measurement



in induction hobs. This sensor uses optical filtering to measure the cooking vessel's base temperature through the ceramic hob surface, improving cooking precision and safety. Traditional induction hobs measure glass temperature, causing inaccuracies. The MLX90617 directly measures cookware temperature, correcting for glass emissions with an advanced algorithm. The company claims that it works with another non-contact temperature sensor mounted below the ceramic surface of the hob. An LED and light sensor can also be used for better accuracy. The sensor provides real-time temperature readings of the cookware, enhancing cooking precision and enabling safer, more

efficient automated cooking systems.
Melexis
<https://www.melexis.com>

Automotive UWB SoC for secure access

Qorvo has introduced the QPF5100Q, an automotive-qualified ultra-wideband (UWB) system-on-chip (SoC). The SoC meets industry demands for accurate and reliable UWB technology in applications like secure keyless entry, digital key, child presence detection, and motion sensing. It offers advanced



UWB functionality and configurable software, allowing automotive designers to enhance performance and differentiate their applications. It features low-power operation and high integration, helping customers future-proof their designs. The SoC allows locking and unlocking without a physical key. It features a motion sensor to detect its owner and functions as a radar, providing distance measurements with 10cm accuracy. Additionally, it prevents relay attacks for enhanced security.

Qorvo
<https://www.qorvo.com>

First UWB radar SoC with 3D beamforming

ARIA Sensing has launched Hydrogen, an ultra-wideband (UWB) radar system-on-chip (SoC) that is the first to offer 3D beamforming with a programmable bandwidth of up to 1.8GHz. The SoC enables 1D, 2D, and 3D detection with radar waveforms, making it suitable for sensing presence, position, vital signs, and gestures while maintaining low power consumption. Hydrogen supports